

Technology Selection and Rationale

DIRECTIONAL · AS AT 3 SEPTEMBER 2026

Directional. First-pass selections for V3, not settled decisions. This view answers which technologies the platform uses and why: the problem each one solves, the features actually relied on, and what was rejected. It deliberately draws no relationships — **LB-V3-L2-01** owns how the platform is constructed. The argument for each selection lives in its ADR; where no V3 ADR exists the card says so, and that gap is a to-do rather than a defect.

Vue 3 + PrimeVue PROVEN IN LAB Experience channel – role-shaped workspaces SOLVES Several role-shaped workspaces must look and behave as one product without one design system per team. RELIES ON Composition API, Pinia stores as client-side read models, PrimeVue v4 theming preset, vue-I18n INSTEAD OF React plus a component library; a hand-rolled design system DECISION no V3 ADR yet USED BY App Shell + MFE; Operator + tenant admin; Driver mobile	Module Federation PROVEN IN LAB Runtime composition of micro-frontends SOLVES A team must ship a workspace feature without rebuilding or redeploying the shell. RELIES ON Runtime remote loading, shared dependency singletons, per-remote versioning INSTEAD OF Build-time monorepo bundling; iframes DECISION no V3 ADR yet USED BY App Shell + MFE; Web delivery (shell + MFE registry)	WorkOS / enterprise IdP PROPOSED Identity federation – authentication only SOLVES Enterprise customers bring their own single sign-on, and Linebooker must not hold their passwords. RELIES ON SSO and directory sync, OIDC and SAML connections, hosted login INSTEAD OF Self-hosted Keycloak; bespoke SSO per tenant DECISION no V3 ADR yet; commercial selection outstanding USED BY Identity federation; Identity + policy broker	OpenFGA PROPOSED Relationship authorisation – who may act on what SOLVES Authority is relational, not a role name: a user may act on one organisation, one depot or one load, and V2 could not express that. RELIES ON Zanzibar-style relationship tuples, an authorisation model per tenant, batch check and list-objects queries INSTEAD OF Role enums in the database; CASL rules duplicated in the frontend; per-service bespoke checks DECISION no V3 ADR yet; self-hosted or hosted not decided USED BY Identity + policy broker; every domain service	SCOPE This is an inventory, not a topology. It answers what and why. LB-V3-L2-01 answers how the parts fit together; the two sit at L2 because they ask different questions. STATUS One decision is accepted for V3. Only ADR-053 is V3-scoped. Thirteen of the sixteen cards carry no V3 ADR at all, and two cite a lab-scoped ADR that does not accept the technology for V3. ADR The ADR is the record, not this sheet. Each card gives one line of reasoning and a pointer. Do not restate an argument here; it will drift and the diagram will win. LAB Proven in lab is not chosen. The Dictionary POC exercised NATS, JetStream, KV, Temporal, Go and the frontend stack at demo scale, under one operator, with no regulated data. GAP Read the amber text as a work list. Each "no V3 ADR yet" marks a selection the platform already leans on without a written decision behind it. STORES Four stores, four jobs. Postgres is transactional truth, JetStream is durable fact history, KV is a derived read model, and object storage holds evidence documents. They are not interchangeable. VENDOR No cloud vendor is selected. S3-compatible names an API contract that several vendors implement. Compute, network and managed-service vendors remain open decisions. DIVERGE Two lab decisions do not carry into V3 as written. ADR-048 chose NATS Object Store for lab documents; V3 proposes an S3-compatible repository instead. ADR-051 is scoped to one lab service. DERIVED Versions, sizing and configuration are out of scope. Pinned versions, high-availability topology, runbooks, subjects and schemas are L3 and L4 decisions. REVIEW A directional document expires. Re-date it whenever a card changes. A position with no date cannot be trusted a year from now.
Go PROVEN IN LAB Domain service implementation language SOLVES Services must start fast and run cheaply as many replicas in every cell. RELIES ON Static binaries, first-class NATS and Temporal SDKs, goroutine concurrency, hexagonal layout INSTEAD OF Java with Spring; Node.js DECISION no V3 ADR yet USED BY Every tile in the domain service plane; service replicas per cell	NATS Core PROVEN IN LAB Service-to-service messaging SOLVES Services must call each other and announce change without point-to-point HTTP wiring or a service mesh. RELIES ON Subject taxonomy, request/reply, queue groups, accounts as hard isolation boundaries INSTEAD OF gRPC with a service mesh; REST between services DECISION no V3 ADR yet USED BY NATS Core + JetStream; NATS provisioning; NATS logical HA per cell	NATS JetStream PROVEN IN LAB Durable fact and event history SOLVES Some state must be reconstructable and auditable, not only current. RELIES ON LimitsPolicy retention so history can be replayed, durable consumers, one domain per cell INSTEAD OF Kafka; an outbox table in every service database DECISION no V3 ADR yet USED BY NATS Core + JetStream; JetStream ZA, AU and BW	NATS KV PROVEN IN LAB Derived read model and fast lookup SOLVES Hot reads must not reach Postgres, and clients need to watch a value for change. RELIES ON Watch, context-scoped keys, write-through from the handler that also writes Postgres INSTEAD OF Redis; caches held inside each service process DECISION no V3 ADR yet USED BY Projections + NATS KV	
Temporal PROVEN IN LAB Long-running process coordination SOLVES Tender expiry, execution, proof, claims and settlement run for days and span many services. RELIES ON Durable timers, automatic retry, workflow state as truth, deterministic replay INSTEAD OF Saga logic written into each service; cron with an outbox DECISION ADR-047, lab scope only; no V3 ADR yet for V3 USED BY Temporal workflows; Temporal runtime per cell	PostgreSQL CHOSEN Transactional source of truth SOLVES Business rules need ACID transactions and a queryable current state owned by one service. RELIES ON ACID transactions, one database and one role per service, JSONB, shared instance by default INSTEAD OF A dedicated instance per service; one shared schema for all services DECISION ADR-052 and ADR-053; ADR-053 is V3 scope USED BY Service-owned Postgres; Service data per cell	S3-compatible object storage PROPOSED Evidence document repository SOLVES Proof of delivery, licences and certificates are large files that must not travel through messaging. RELIES ON The S3 API contract, presigned URLs, a bucket per cell, lifecycle rules INSTEAD OF NATS Object Store as the default; document blobs held in Postgres DECISION ADR-048 chose NATS Object Store for the lab; V3 differs and has no V3 ADR yet USED BY Document store adapter; Evidence store per cell	OpenTelemetry / OTLP PROVEN IN LAB Telemetry contract for logs, metrics and traces SOLVES Cells run independently, so telemetry must be collected the same way everywhere and stay in region. RELIES ON The OTLP wire format, vendor-neutral SDKs, masking at the collector INSTEAD OF A vendor agent embedded in every service DECISION no V3 ADR yet USED BY Observe + audit; Resident operations per cell	
ULID PROVEN IN LAB Entity identity format SOLVES Identifiers appear in NATS subjects and KV keys, so they must be subject-safe and time-sortable. RELIES ON 26 Crockford base32 characters, lexicographic time ordering, minted in application code INSTEAD OF UUIDv4; identifiers generated by the database DECISION ADR-051, lab scope only; no V3 ADR yet for V3 USED BY organizations-service identity; every subject and KV key token	Container platform OPEN Where services actually run SOLVES Services must be packaged and scheduled the same way in every region cell, with room to run a sovereign cell separately. RELIES ON Nothing yet; the lab runs Docker Compose on one host INSTEAD OF V2 ran the API directly on Compute Engine virtual machines with no orchestrator DECISION no V3 ADR yet; Kubernetes, a managed container runtime and plain VMs are all still open USED BY Every deployed element in LB-V3-L2-01 sheet 2	Cloud provider OPEN The infrastructure the cells sit on SOLVES Regional cells and data residency rules must be satisfied by real regions in a real provider. RELIES ON Nothing yet; only the S3 and OTLP API contracts are assumed portable INSTEAD OF V2 is wholly on Google Cloud in one project DECISION no V3 ADR yet; a deliberate open decision, not an oversight USED BY Every regional cell; object storage; managed data services	Build and release pipeline OPEN How code reaches a cell SOLVES Independently deployed micro-frontends and services need independent, repeatable release paths. RELIES ON Nothing yet; the lab builds with Vite and Go toolchains directly INSTEAD OF V2 uses Google Cloud Build with fifteen pipelines, plus GitHub Actions for notifications DECISION no V3 ADR yet USED BY App Shell + MFE registry; every service	
How to read a card. The left edge colours the plane the technology serves, using the same colour meanings as LB-V3-L2-01. The chip states the selection, not the document: Chosen is accepted for V3 with an ADR of record, Proposed is a directional position with no V3 ADR yet, and Proven in lab was exercised in the Dictionary POC but is not yet accepted for V3. Used by names the LB-V3-L2-01 elements that depend on the selection. No cloud vendor is listed: object storage is an API contract here, not a product choice.				

KEY

Blue edge — experience and entry plane	Purple edge — domain service plane	Cyan edge — platform foundations	Green edge — cross-cutting platform concern
CHOSEN Accepted for V3, with an ADR of record	PROPOSED Directional position; no V3 ADR yet	PROVEN IN LAB Exercised in the Dictionary POC; not yet accepted for V3	no V3 ADR yet A selection with no written V3 decision behind it
USED BY The LB-V3-L2-01 elements that depend on this selection	realises The role the technology fills, stated before the product	no lines An inventory view; relationships are drawn in LB-V3-L2-01	DIRECTIONAL A first-pass position, dated on the sheet, not a settled decision

ACRONYMS

ACID Atomic, consistent, isolated, durable	POC Proof of concept
ADR Architecture decision record	S3 The object-storage API; S3-compatible means any vendor implementing it
API Application programming interface	SAML Security Assertion Markup Language
HA High availability	SDK Software development kit
IdP Identity provider	SSO Single sign-on
JSONB PostgreSQL binary JSON column type	ULID Universally unique lexicographically sortable identifier
KV Key/value store (NATS KV)	UUID Universally unique identifier
MFE Micro-frontend	ZA / AU / BW South Africa / Australia / Botswana (ISO country codes)
NATS The messaging system; a product name, not an abbreviation	
OIDC OpenID Connect	
OTLP OpenTelemetry protocol	

External Services and Integrations

DIRECTIONAL · AS AT 3 SEPTEMBER 2026

Directional. Third-party services the platform depends on, carried forward from V2 unless stated. Sheet 1 covers technologies Linebooker builds on; this sheet covers services Linebooker buys or integrates with, where the vendor owns the roadmap and the outage. Every card here is inherited operational reality, not a fresh V3 selection — none has been re-tendered, and the final card records what is deliberately not carried forward.

Acumatica PROPOSED Accounting and finance system of record SOLVESInvoices, debtors and customer master data live in a finance system Linebooker does not own. RELIES ONREST contract API, OAuth token exchange, customer and vendor entity sync INSTEAD OFBuilding invoicing inside the platform; the live Google Sheet V2 still reads debtors from DECISIONno V3 ADR yet; carried from V2, replacement not assessed USED BYFinancial architecture; billing and settlement	monday.com PROPOSED Commercial pipeline and account boards SOLVESSales and account teams work customers and transporters outside the operational platform. RELIES ONBoard and item API, item identifiers stored against transporters INSTEAD OFA customer-relationship module inside Linebooker DECISIONno V3 ADR yet; carried from V2, a candidate to retire USED BYOperator workspace; participant onboarding	Meta WhatsApp Cloud API PROPOSED Conversational channel to drivers and transporters SOLVESTransporters and drivers work from a phone and will not install or log in to a portal for every update. RELIES ONCloud API graph endpoints, message templates, inbound webhooks INSTEAD OFNative mobile push alone; SMS for every interaction DECISIONno V3 ADR yet; carried from V2, load-bearing in V2 USED BYDriver mobile; notification fan-out	SendGrid PROPOSED Transactional email delivery SOLVESThe platform must send templated, deliverable email at volume and prove it was sent. RELIES ONTemplate API, event webhooks, domain authentication INSTEAD OFSelf-hosted SMTP; the Spring mail fallback V2 still carries DECISIONno V3 ADR yet; carried from V2, vendor-swappable USED BYNotification fan-out; document distribution
Twilio Verify PROPOSED Second factor over SMS and voice SOLVESUsers without an enterprise identity provider still need a second factor. RELIES ONVerify service, one-time passcode issue and check INSTEAD OFRolling one-time passcodes in-house; authenticator app only DECISIONno V3 ADR yet; carried from V2; overlaps WorkOS multi-factor, one should go USED BYIdentity federation; Identity + policy broker	Peach Payments PROPOSED Card acceptance in South Africa SOLVESCustomers must pay by card without card data touching Linebooker systems. RELIES ONHosted checkout, entity identifiers per brand, status callbacks INSTEAD OFHolding card data; a global-only processor with no local acquiring DECISIONno V3 ADR yet; carried from V2; region-specific, not portable to AU USED BYFinancial architecture; payment capture	Ozow PROPOSED Instant bank transfer in South Africa SOLVESMany South African customers pay by electronic funds transfer, not card. RELIES ONPayment request API, transaction status polling INSTEAD OFManual proof-of-payment reconciliation DECISIONno V3 ADR yet; carried from V2; region-specific, not portable to AU USED BYFinancial architecture; payment capture	Telematics broker PROPOSED One tracking contract over many vendors SOLVESAround thirty-eight telematics vendors each expose a different feed, and a transporter may use any of them. RELIES ONPer-transporter vendor credentials, normalised position and event stream INSTEAD OFA direct integration per vendor inside each domain service DECISIONno V3 ADR yet; carried from V2 as Linetracker; V3 shape not designed USED BYTracking and visibility; load execution
Google Maps Platform PROPOSED Geocoding, distance and routing SOLVESAddresses must resolve to coordinates and legs must be priced and timed by real road distance. RELIES ONGeocoding, distance matrix and directions APIs INSTEAD OFOpen routing engines self-hosted; straight-line distance DECISIONno V3 ADR yet; carried from V2; cost scales with volume USED BYPricing; route planning; operating areas	MapLibre GL + MapTiler PROPOSED Map rendering in the browser SOLVESOperators need interactive maps of operating areas and live positions without a per-view licence cost. RELIES ONVector tile serving, self-hosted boundary tiles, open rendering library INSTEAD OFRendering every map through the Google Maps JavaScript API DECISIONno V3 ADR yet; carried from V2; splits map cost from map data USED BYOperator workspace; operating areas	Google Document AI PROPOSED Extraction from scanned evidence SOLVESProof of delivery arrives as a photograph or scan and must become structured data. RELIES ONA trained processor, batch and online extraction INSTEAD OFManual capture; a generic optical character reader with no document model DECISIONno V3 ADR yet; carried from V2; the strongest single cloud lock-in USED BYEvidence store; proof of delivery workflow	BigQuery + Looker Studio PROPOSED Analytics warehouse and reporting SOLVESOperational stores must not serve analytics, and customers ask for their own reports. RELIES ONWarehouse dataset, signed embedded reports INSTEAD OFReporting straight off the transactional database DECISIONno V3 ADR yet; carried from V2; a projection target, never a source of truth USED BYFinancial architecture; customer reporting
Customer file drops PROPOSED Contracted file exchange with large shippers SOLVESSome enterprise customers accept documents only over their own secure file transfer endpoint. RELIES ONSecure file transfer client, per-customer credentials and folder rules INSTEAD OFEmail attachments; asking the customer to use the portal DECISIONno V3 ADR yet; carried from V2 as a standalone middleware service USED BYExternal integration; document distribution	Voucher rewards PROPOSED Driver and transporter incentives SOLVESDrivers earn rewards that must be redeemed for something they actually want. RELIES ONRetailer voucher denominations and redemption INSTEAD OFCash incentives; a points scheme with no redemption path DECISIONno V3 ADR yet; carried from V2; single retailer, region-specific USED BYDriver mobile; rewards	reCAPTCHA PROPOSED Bot protection on public forms SOLVESPublic sign-up and enquiry forms are scripted against. RELIES ONServer-side token verification INSTEAD OFRate limiting alone DECISIONno V3 ADR yet; carried from V2; a candidate to fold into the edge USED BYPublic web entry; participant onboarding	Not carried into V3 NOT CARRIED V2 technologies deliberately left behind SOLVESA rewrite is the moment to stop paying for scaffolding no one uses. RELIES ONNothing; listed so the omission is visible rather than accidental INSTEAD OFJHipster scaffolding, Spring Boot 2.1, MySQL, RabbitMQ, Redis cache, Elasticsearch, Logstash, Dropwizard metrics, React 17 + Redux + Material UI, CASL, jOOQ, Liquibase DECISIONDirectional: each of these has a V3 replacement on sheet 1 or is simply dropped USED BYNothing; this card exists to close the inventory
How to read a card. The amber left edge marks an external or separately governed service — the same colour meaning as LB-V3-L2-01. Carried from V2 means the dependency exists in the running platform today and has not been reassessed for V3; it is a statement of fact, not an endorsement. Used by names the LB-V3-L2-01 elements that would depend on the service.			

SCOPE	Bought, not built. Sheet 1 lists what the platform is constructed from. This sheet lists services outside the trust and release boundary, where an outage is somebody else's incident.
SOURCE	Every card was read out of the V2 codebase. This is an observed inventory of the running platform, not a wish list. Absence from this sheet means absence from V2.
STATUS	No card here is chosen for V3. Carrying a vendor forward by default is itself a decision; it has simply not been written down yet.
REGION	Four services are South Africa only. Card payments, bank transfer, the voucher retailer and most telematics vendors do not extend to an Australian cell. Each needs a regional equivalent or a deliberate gap.
LOCK-IN	Document extraction is the deepest lock-in. A trained processor is not portable between vendors. Payments, email and messaging are comparatively swappable.
OVERLAP	Two services do the same job. The identity provider and the verification service both offer a second factor. Keeping both is a cost and a support burden.
CONTRACT	Integrate against a contract, never a vendor. Each of these belongs behind an adapter owned by one service, so a vendor change is one adapter, not a migration.
DROPPED	The last card is the honest half. An inventory that lists only what is kept hides the size of the change. What is dropped is a decision that needs a record too.
SECRETS	Every service on this sheet holds a credential. V3 must source them from a secret store; this sheet does not name endpoints, keys or account identifiers.
NEXT	L3-04 External Integration owns the detail. Adapter shape, retry and idempotency rules, and per-vendor contracts belong there, not on this sheet.

KEY

	Amber edge — external or separately governed service		Grey edge — not a selection; a record of what is dropped	PROPOSED	Directional position; no V3 ADR yet	NOT CARRIED	Present in V2 and deliberately left behind
carried from V2	Runs in the platform today; not reassessed for V3	no V3 ADR yet	A dependency with no written V3 decision behind it	USED BY	The LB-V3-L2-01 elements that depend on the service	realises	The role the service fills, stated before the vendor
no lines	An inventory view; relationships are drawn in LB-V3-L2-01	DIRECTIONAL	A first-pass position, dated on the sheet, not a settled decision	no vendors	No endpoint, key or account identifier appears on this sheet	V2	The Linebooker platform running in production today

ACRONYMS

ADR	Architecture decision record	OCR	Optical character recognition
API	Application programming interface	POD	Proof of delivery
AU	Australia (ISO country code)	REST	The HTTP resource API style
BI	Business intelligence	SFTP	Secure file transfer protocol
CRM	Customer relationship management	SMS	Short message service
EFT	Electronic funds transfer	V2 / V3	The current and proposed Linebooker platform versions
ERP	Enterprise resource planning	ZA	South Africa (ISO country code)
IdP	Identity provider		
OAuth	The delegated authorisation standard		